Islamic University of Technology (IUT)
Organization of Islamic Cooperation (OIC)

Winter Semester (Academic Year: 2023-2024)

Course Title: Computational Methods in Engineering Lab

Assignment Statement

Nano-magnetism:

Nano-magnetism is a field that studies the magnetic properties of materials at the nanometer scale. Magnetic phenomena at this scale can be quite different from those in bulk materials, due to the dominance of surface and edge effects and quantum mechanical effects. The magnetic behavior of small particles or thin films is particularly important for applications in data storage, spintronics, and magnetic sensors.

The theoretical understanding of magnetization dynamics in ferromagnetic materials was significantly advanced by Lev Landau and Eugen Lifshitz in the 1930s with their formulation of the Landau-Lifshitz (LL) equation, which describes the time evolution of magnetization under external fields. In the 1950s, Gilbert extended this theory by introducing a damping term to account for energy dissipation, resulting in the Landau-Lifshitz-Gilbert (LLG) equation, a cornerstone of micromagnetic modeling. Later, in 1996, Slonczewski further expanded this framework by incorporating spin-transfer torque (STT) effects, enabling the description of magnetization dynamics under spin-polarized currents. Together, these developments have shaped our understanding of precession, relaxation, and current-driven switching in magnetic materials.

Mathematical LLGS equation is expressed as:

$$(\mathbf{1}+\boldsymbol{\alpha}^{\mathbf{2}})\frac{d\hat{\boldsymbol{m}}}{dt} = -\boldsymbol{\gamma}\left(\hat{\boldsymbol{m}}\times\hat{\boldsymbol{H}}_{\boldsymbol{eff}}\right)-\boldsymbol{\alpha\gamma}\left(\hat{\boldsymbol{m}}\times(\hat{\boldsymbol{m}}\times\hat{\boldsymbol{H}}_{\boldsymbol{eff}}\right) \quad \text{.....(1)}$$

Here,

- $\hat{m}$ is the unit magnetization vector
- $\hat{H}_{eff}$ is the effective magnetic field containing all internal and external fields acting on the nanomagnet
- α is the Gilbert damping factor and the usually very small value for CoFeB is 0.01.
- γ is the gyromagnetic ratio for electrons and expressed as $\boldsymbol{\gamma} = \frac{\mathbf{2}\boldsymbol{\mu_b}}{\hbar}$

The first term on the right-hand side of the LLG equation, represents the precession torque. This term accounts for the rotation of the magnetization $\hat{\boldsymbol{m}}$ around the effective field $\boldsymbol{H_{eff}}$. The cross-product $\hat{\boldsymbol{m}}\times\boldsymbol{H_{eff}}$ implies that the magnetization vector precesses around the direction of the effective field. The constant $\boldsymbol{\gamma}$ is the **gyromagnetic ratio**, a fundamental material constant that relates the magnetization to the magnetic field.

The effective field, $\boldsymbol{H_{eff}}$ is the sum of all the fields that influence magnetization dynamics of nanomagnet. For this assignment we can consider, $\boldsymbol{H_{eff}}$ includes,

$$\boldsymbol{H_{eff}} = \boldsymbol{H_k} + \boldsymbol{H_{ext}}$$

Where,

- **Anisotropy field ($\boldsymbol{H_k}$)**: This arises from the magnetic anisotropy of the material, which favors alignment of the magnetization in certain directions (e.g., along a crystal axis). So, it depends on the direction. For this assignment we will consider $\boldsymbol{H_k}$ is in z direction. Along the z axis it can be +z or -z. To simplify we can write $\hat{H}_k = H_k\, m_z\, \hat{z}$, where m_z is the z-component of magnetization $\hat{\boldsymbol{m}}$,

and $\hat{z}$ the unit vector along the z-direction. Since the anisotropy field $\boldsymbol{H_k}$ is here in z direction, so the stable position of the nanomagnet is ±z direction ($\hat{\boldsymbol{m}} = (\boldsymbol{0}, \boldsymbol{0}, \pm\boldsymbol{1})$).

- **External magnetic field ($\boldsymbol{H_{ext}}$)**: This is the external applied magnetic field, which drives the magnetization in a particular direction.

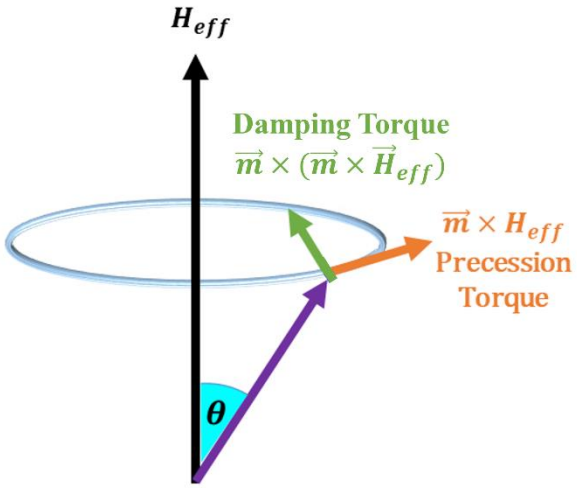


Figure: Magnetization Dynamics of according to LLGS equation.

The second term in the LLGS equation, represents the **damping torque**. The damping term accounts for the loss of energy due to the interaction of the magnetic system with its surroundings (such as the lattice or thermal bath). The **Gilbert damping constant** (α) quantifies the strength of this damping. It is typically a small number, on the order of 0.001 to 0.1, depending on the material.

The damping term causes the magnetization to relax towards the direction of the effective field. This process is responsible for the dissipation of energy and the eventual stabilization of the magnetization in a steady-state configuration, especially after the magnetization has precessed around the effective field.

For easier solving, we can normalize the **equation (1)** by dividing it with $\boldsymbol{\gamma H_k}$, which results in the following equation;

$$\frac{d\hat{\boldsymbol{m}}}{d\boldsymbol{\tau}} = -\left(\hat{\boldsymbol{m}} \times \hat{\boldsymbol{h}}\right) - \boldsymbol{\alpha}(\hat{\boldsymbol{m}} \times (\hat{\boldsymbol{m}} \times \hat{\boldsymbol{h}})$$

Where, $\boldsymbol{\tau} = \frac{t\gamma \boldsymbol{H_k}}{1+\boldsymbol{\alpha}^2}$ and $\hat{h} = \frac{\hat{H}_{ext}}{H_k} + m_z\,\hat{z}$.

Read wiki about different terms for more details.

Pure Relaxation: When the magnetization is in a non-equilibrium position ($\hat{\boldsymbol{m}} \neq (\boldsymbol{0}, \boldsymbol{0}, \pm\boldsymbol{1})$), and only $\boldsymbol{H_k}$ is present (with no external field, $\boldsymbol{H_{ext}}$), it relaxes along a precessional path toward an equilibrium position, either +z or -z, driven by the damping torque.

A sample code for a scenario is given where the LLG equation is solved using **ode45** module of MATLAB. The simulation parameter used in the code are as follows;

Damping Factor, α	0.01
Gyromagnetic Ration, γ	2.2127×10^5 m/(A.s)
Anisotropy Field, H_k	3.1951×10^5 A/m

```
clc;
clear all;
close all;

global alpha;
alpha = 0.01;
gamma = 2.21276e5;     %gyromagnetic ratio
mu = 1.2566e−6;          %permeability in vacuum.
m_initial = [0.01 0.95 0.01];

m_initial = m_initial/norm(m_initial);
h_k = 3.1951e5;

%solving LLG equation for pure relaxation
[tau, m_solve]=ode45(@pure, [0 , 2000] , m_initial) ;

%converting it in the time domain
time = tau*(1+alpha^2) / (gamma *h_k );

subplot(3,1,1)
plot(time*1e9,m_solve(:,1));
xlabel('time (ns)')
ylabel('m_x')
ylim([ min(m_solve(:,1)) max(m_solve(:,1))])
title(['m_x over time (initial value of m_x= ',num2str(m_initial(1)),')'])
subplot(3,1,2)
plot(time*1e9,m_solve(:,2));
xlabel('time (ns)')
ylabel('m_y')
title(['m_y over time (initial value of m_y= ',num2str(m_initial(2)),')'])
ylim([ min(m_solve(:,2)) max(m_solve(:,2))])
subplot(3,1,3)
plot(time*1e9,m_solve(:,3));
xlabel('time (ns)')
ylabel('m_z')
title(['m_z over time (initial value of m_z= ',num2str(m_initial(3)),')'])
ylim([ min(m_solve(:,3)) max(m_solve(:,3))])
```

```
function llg = pure (tau, m)
H = [ 0 , 0 , m(3)];
global alpha
dynamic = cross (m, H);
damping = cross (m, dynamic);

llg = [ − dynamic (1) − alpha*damping (1) ; − dynamic(2) − alpha*damping(2) ;
        − dynamic (3) − alpha*damping(3) ] ;

end
```

Output:

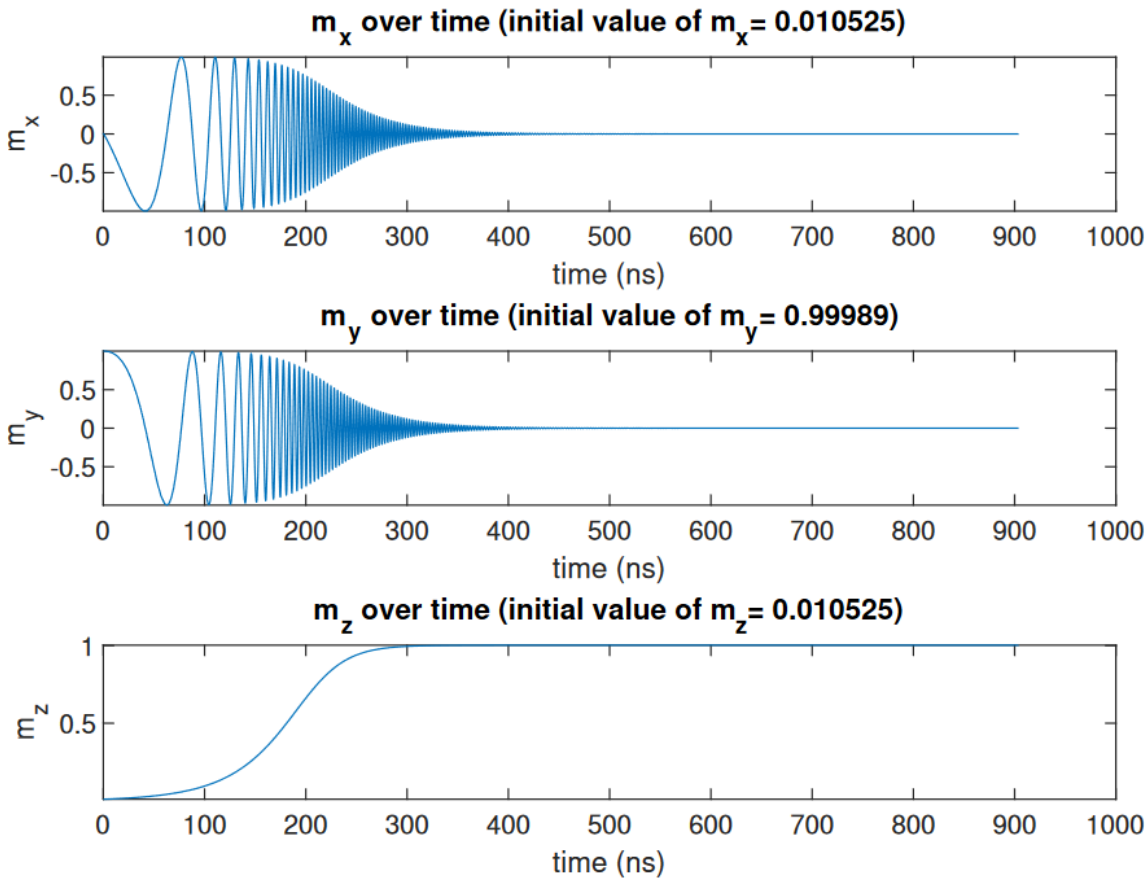


Fig: Evolution of $\hat{m} = (m_x, m_y, m_z)$ with time.

The initial condition was set **to $\hat{m}$= (0.010525, 0.99989, 0.010525),** so initially $\hat{m}(0)$ was along the y-axis. But ultimately it settled along +z-axis aligning with the H_k field.

Task 01: Write a MATLAB code to solve the magnetization, $\hat{m} = (m_x, m_y, m_z)$ with time for the example pure relaxation scenario shown above. **You can not use ode45 module**, instead implement a numerical solver.

Task 02: Vary the initial magnetization position $\hat{\boldsymbol{m}}(\mathbf{0})$**,** and observe the time required for the magnetization to reach a stable equilibrium state (+z or -z). Present your observations in a tabular format, and provide appropriate comments on how the initial position affects the relaxation time.

Task 03: Investigate the effect of the Gilbert damping factor (α) on the relaxation dynamics by varying α in the range 0.001 to 0.5 (use at least 10 values). For a fixed initial magnetization positions (not a stable state), observe how α influences the time required to reach equilibrium. Present your results in a tabular format, and provide detailed comments on the relationship between damping and relaxation time.

Submission:

You need to show the code physically to a faculty on the 13th week along with the report at your respective class time. Once the faculty approve your submissions, you will upload the code and report in the google classroom.

Submitting code and report without showing any faculty is not accepted.